Class Exercises: Arithmetic & Geometric Sequences

LEVEL 2: MEDIUM (Algebraic Parameters & Structural Patterns)

Exercise 1 (Medium - Arithmetic)

An arithmetic sequence has its 4th term equal to 17 and its 9th term equal to 42.

a) Find the first term a and the common difference d.

b) Determine which term in the sequence is equal to 102.

Exercise 2 (Medium - Geometric)

The third term of a geometric sequence is 12 and the sixth term is 96.

a) Find the first term a and the common ratio r.

b) Calculate the exact value of the 10th term u_{10}.

Exercise 3 (Medium - Algebraic Terms)

The consecutive terms $k-1$, $2k+1$, and $5k+3$ form a geometric sequence.

a) Show that $k^2 - 6k - 4 = 0$ or find the possible value(s) of k.

b) Assuming $k > 0$, determine the exact value of the common ratio r.

Exercise 4 (Medium - Sierpinski Fractal)



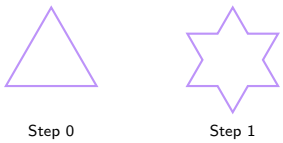
Step 0 Step 1 Step 2

A Sierpinski Triangle starts with 1 solid black equilateral triangle at Step 0. At each step, the central inverted triangle is removed, creating 3 smaller black triangles.

a) Write down the number of black triangles N_n for Steps 0, 1, 2, and 3.

b) State the n-th term formula for N_n and calculate the number of black triangles at Step 8.

Exercise 5 (Medium - Koch Snowflake Perimeter)



Step 0 Step 1

The Koch Snowflake starts at Step 0 with an equilateral triangle of perimeter $P_0 = 3$ cm. At each iteration, the middle third of every segment is replaced by two segments of equal length.

a) Express the perimeter P_n after n iterations as a geometric sequence.

b) Calculate the total perimeter P_5 correct to 2 decimal places.

Exercise 6 (Medium - Consecutive Sums)

The sum of three consecutive terms in an arithmetic sequence is 33, and their product is 1215.

a) Express the three terms as $x - d$, x, and $x + d$ to find x.

b) Find the two possible values for the common difference d and list the sequence terms.

LEVEL 3: HARD (Simultaneous Equations, Fractals & Advanced Modeling)

Exercise 7 (Hard - Mixed Sequence)

The first, third, and seventh terms of an arithmetic sequence with common difference $d \neq 0$ form the first three consecutive terms of a geometric sequence.

a) Show that the first term a is equal to $2d$.

b) Find the exact value of the common ratio r of the geometric sequence.

Exercise 8 (Hard - Cantor Set Area/Length)

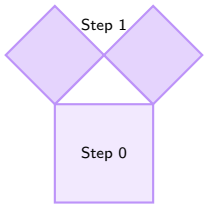


The Cantor Set starts at Step 0 with a line segment of length $L_0 = 1$ m. At each step, the open middle third of every remaining interval is removed.

a) Find the total remaining length L_n after n steps.

b) Determine after how many steps n the total length first falls below 0.01 m (1%).

Exercise 9 (Hard - Pythagoras Tree Squares)



A Pythagoras Tree fractal starts at Step 0 with 1 square of area $A_0 = 16$ cm^2. At Step 1, two right isosceles triangles build 2 smaller squares on top, each having half the area of the parent square.

a) Find the total area T_n of all new squares added at Step n.

b) Find the accumulated area of all squares combined up to Step 6.

Exercise 10 (Hard - Financial Depreciation / Compound Growth)

A industrial machine worth $\$50,000$ depreciates in value by 12% each year.

a) Express its value V_n at the end of n years as a geometric sequence.

b) Find the minimum number of complete years for the machine's value to drop below $\$15,000$.

Detailed Solutions: Arithmetic & Geometric Sequences

LEVEL 2: MEDIUM SOLUTIONS

Solution 1 (Medium - Arithmetic)

1. Set up simultaneous equations for u_4 and u_9:

$$u_4 = a + 3d = 17$$

$$u_9 = a + 8d = 42$$

2. Subtract u_4 from u_9:

$$5d = 25 \implies \mathbf{d = 5}$$

$$a + 3(5) = 17 \implies \mathbf{a = 2}$$

3. Solve $u_n = 102$:

$$2 + (n-1)5 = 102 \implies 5(n-1) = 100$$

$$n - 1 = 20 \implies \mathbf{n = 21}$$

Solution 2 (Medium - Geometric)

1. Set up terms u_3 and u_6:

$$u_3 = a \cdot r^2 = 12, \quad u_6 = a \cdot r^5 = 96$$

2. Divide u_6 by u_3:

$$\frac{a \cdot r^5}{a \cdot r^2} = \frac{96}{12} \implies r^3 = 8 \implies \mathbf{r = 2}$$

3. Find a:

$$a \cdot 2^2 = 12 \implies 4a = 12 \implies \mathbf{a = 3}$$

4. Compute 10th term:

$$u_{10} = 3 \cdot 2^9 = 3 \cdot 512 \implies \mathbf{u_{10} = 1536}$$

Solution 3 (Medium - Algebraic Terms)

1. Geometric property: ratio of consecutive terms is constant:

$$\frac{2k+1}{k-1} = \frac{5k+3}{2k+1}$$

2. Cross-multiply:

$$(2k+1)^2 = (k-1)(5k+3)$$
$$4k^2 + 4k + 1 = 5k^2 - 2k - 3$$
$$0 = k^2 - 6k - 4$$

3. Use quadratic formula for $k > 0$:

$$k = \frac{6 + \sqrt{36 - 4(1)(-4)}}{2} = \frac{6 + \sqrt{52}}{2} = \mathbf{3} + \sqrt{\mathbf{13}} \approx 6.61$$

4. Compute ratio r:

$$\mathbf{r \approx 2.53}$$

Solution 4 (Medium - Sierpinski Fractal)

1. Count black triangles per step:

$$N_0 = 1, \quad N_1 = 3, \quad N_2 = 9, \quad N_3 = 27$$

2. Identify as Geometric sequence with $a = 1$ and $r = 3$:

$$N_n = 3^n$$

3. For Step 8 ($n = 8$):

$$N_8 = 3^8$$

$$\mathbf{N_8 = 6561} \texttt{ black triangles}$$

Solution 5 (Medium - Koch Snowflake)

1. Each iteration increases segment count by factor of 4 and reduces segment length by $\frac{1}{3}$:

$$P_n = P_0 \cdot \left(\frac{4}{3}\right)^n = 3 \cdot \left(\frac{4}{3}\right)^n$$

2. Compute perimeter at $n = 5$:

$$P_5 = 3 \cdot \left(\frac{4}{3}\right)^5 = 3 \cdot \frac{1024}{243} = \frac{1024}{81}$$

$$\mathbf{P_5 \approx 12.64 \ \texttt{cm}}$$

Solution 6 (Medium - Consecutive Sums)

1. Sum of terms:

$$(x-d) + x + (x+d) = 33 \implies 3x = 33 \implies \mathbf{x = 11}$$

2. Product of terms:

$$(11-d)(11)(11+d) = 1215$$

$$11(121 - d^2) = 1215 \implies 121 - d^2 = \frac{1215}{11} \approx 110.45$$

$$d^2 = 121 - \frac{1215}{11} = \frac{116}{11} \implies \mathbf{d \approx \pm 3.25}$$

3. Terms are approx: $\mathbf{7.75, 11, 14.25}$

LEVEL 3: HARD SOLUTIONS

Solution 7 (Hard - Mixed Sequence)

1. Arithmetic terms: $u_1 = a$, $u_3 = a + 2d$, $u_7 = a + 6d$. 2. Geometric property:

$$\frac{a+2d}{a} = \frac{a+6d}{a+2d} \implies (a+2d)^2 = a(a+6d)$$

$$a^2 + 4ad + 4d^2 = a^2 + 6ad \implies 4d^2 = 2ad$$

3. Since $d \neq 0$, divide by $2d$:

$$2d = a \implies \mathbf{a = 2d}$$

4. Find common ratio r:

$$r = \frac{a+2d}{a} = \frac{2d+2d}{2d} = \frac{4d}{2d}$$

$$\mathbf{r = 2}$$

Solution 8 (Hard - Cantor Set)

1. Total remaining length after n steps:

$$L_n = 1 \cdot \left(\frac{2}{3}\right)^n$$

2. Set inequality $L_n < 0.01$:

$$\left(\frac{2}{3}\right)^n < 0.01$$

3. Take natural logarithms:

$$n \ln\left(\frac{2}{3}\right) < \ln(0.01) \implies n(-0.4055) < -4.6052$$

$$n > \frac{-4.6052}{-0.4055} \approx 11.35$$

$$\mathbf{n = 12 \ \texttt{steps}}$$

Solution 9 (Hard - Pythagoras Tree)

1. Number of new squares at Step n is 2^n. Area of each individual square at Step n is $16 \cdot (1/2)^n$.
2. Total new area added at Step n:

$$T_n = 2^n \cdot \left[16 \cdot \left(\frac{1}{2}\right)^n\right] = 16 \text{ cm}^2$$

(Notice that the total added area is constant at each step!) 3. Accumulated total area up to Step 6 (including Step 0 to Step 6, i.e., 7 layers):

$$\text{Total Area} = 16 \times 7$$

$$\texttt{Total Area} = \mathbf{112}\ \texttt{cm}^2$$

Solution 10 (Hard - Depreciation)

1. Formula for value after n years ($r = 1 - 0.12 = 0.88$):

$$V_n = 50000 \cdot (0.88)^n$$

2. Set inequality $V_n < 15000$:

$$50000 \cdot (0.88)^n < 15000 \implies (0.88)^n < 0.3$$

3. Take logarithms:

$$n \log(0.88) < \log(0.3) \implies n(-0.0555) < -0.5229$$

$$n > \frac{-0.5229}{-0.0555} \approx 9.42$$

$$\mathbf{n = 10}\ \texttt{years}$$